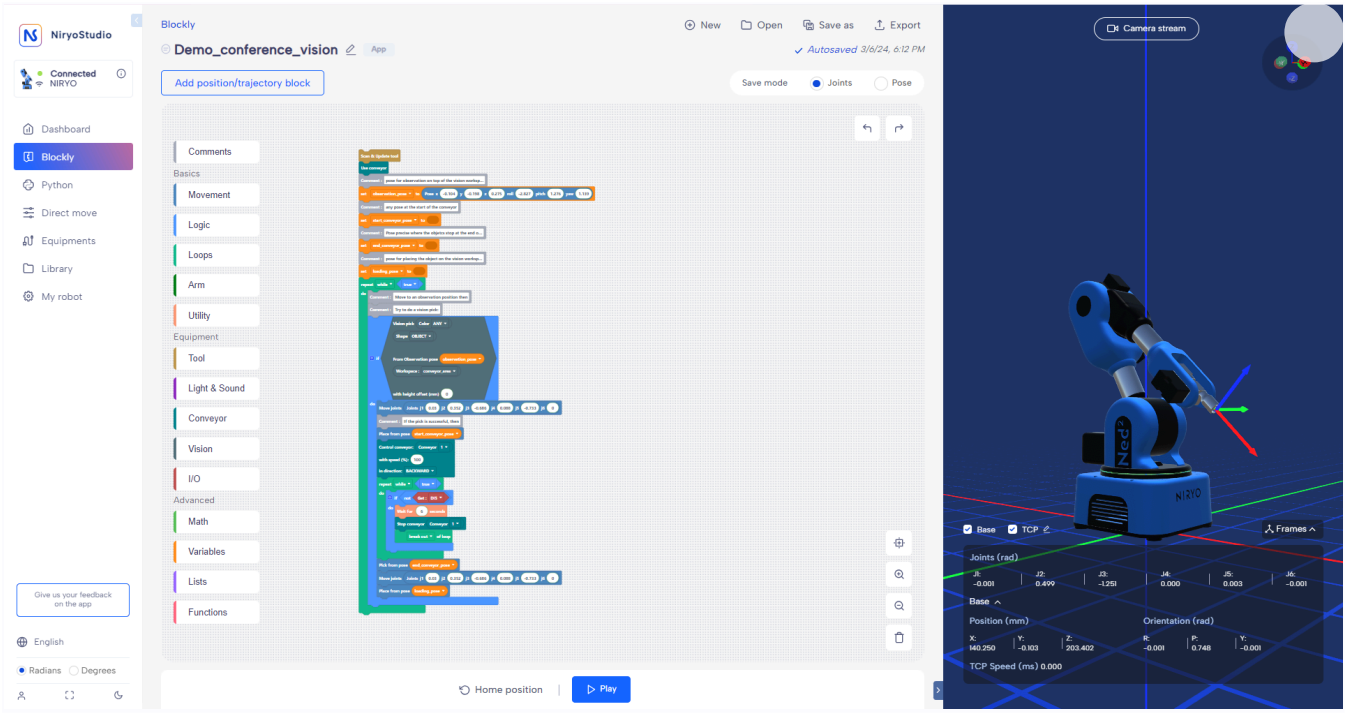
Blockly

Code your first program easily with block

On the Blockly tab, you can easily create your first program by dragging and dropping blocks onto the playground.

For more information, visit Blockly API Documentation.

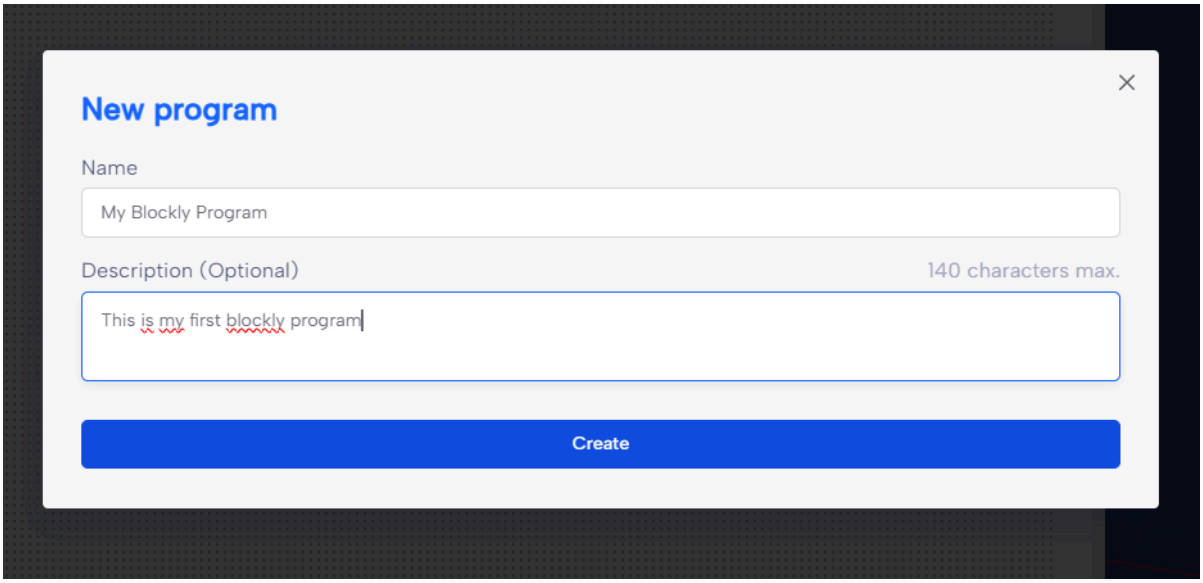
> This section is available even if you are not connected to a robot. You can create or edit Blockly programs, but you cannot run them.



Create a new program

When you enter the Blockly tab, NiryoStudio prompts you to create a new program.
You must enter a unique name for the program. If the name is already in use, NiryoStudio will ask if you want to overwrite the existing program.
You can also add an optional description to help you remember the purpose of the program.

Once everything is set, click **Create**, and a new, empty Blockly program will be created on your computer.



Explore Blockly collection

Blocks are grouped by theme. Click on a theme to open it and view all available blocks. Drag and drop blocks onto the playground.

For more information, consult

Types of Blocks
https://docs.niryo.com/api/blockly/article/#section-header-two-5s53j

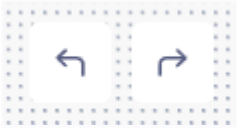
or

Interacting with Blocks
https://docs.niryo.com/api/blockly/article/#section-header-two-d0k0l

to learn how to manage blocks on the playground.

Playgound controls

The Blockly playground has several controls:



- Undo changes in the playground.

- Redo changes in the plaground.



- Center playground on blocks.

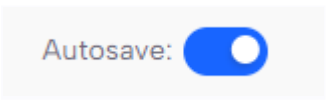
- Zoom in action.

- Zoom out action.

- Delete action.

Save a program

You can save your program manually by clicking the **Save** button, or enable **Autosave** to let NiryoStudio save your program automatically with every change.



Play a program and Home Position

Once you have composed your program, you can test it by clicking the **Play** button. This button is available only when you are connected to a robot.

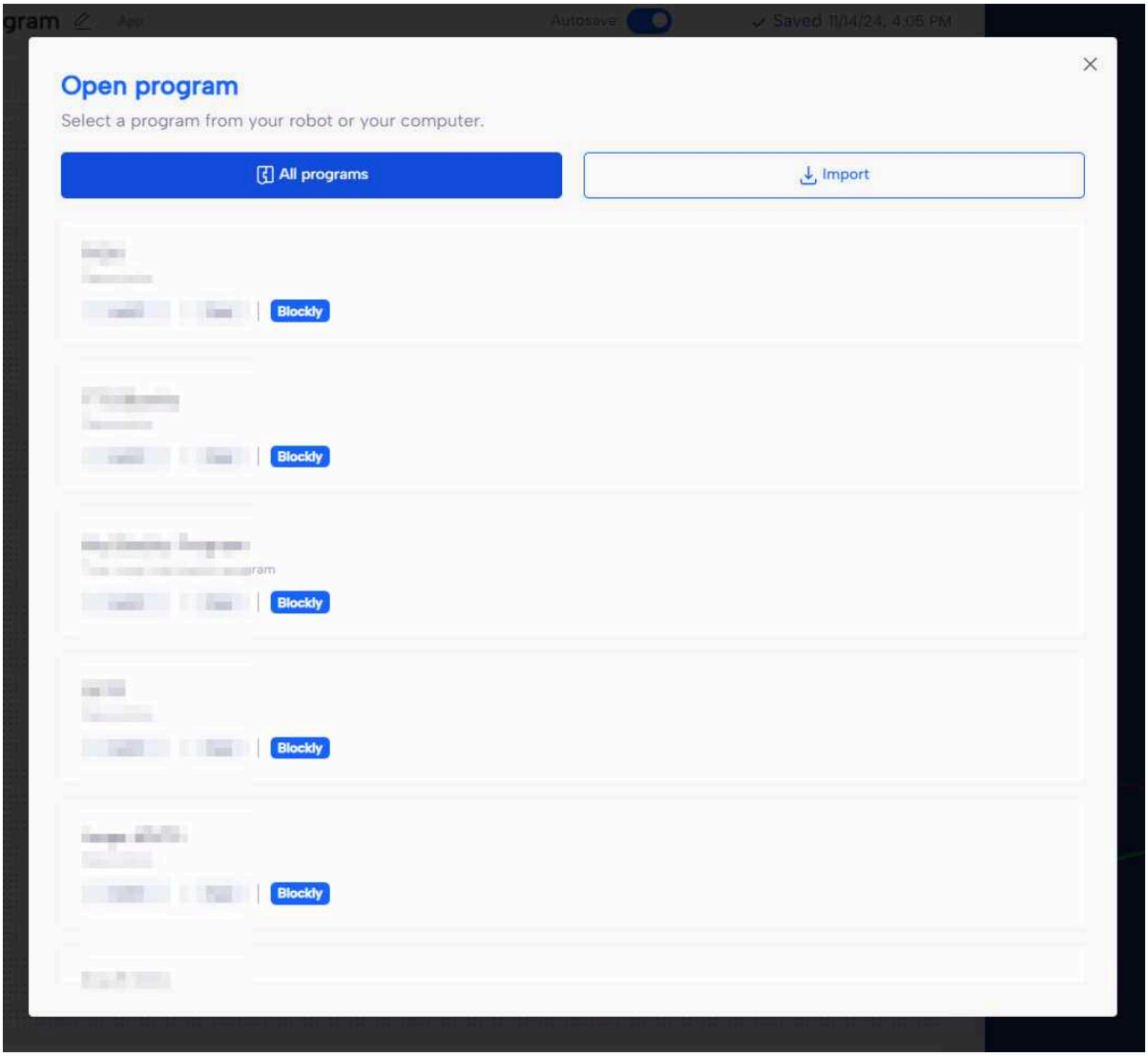
Once execution starts, the "Play" button changes to a **Stop** button, allowing you to interrupt your program by clicking it.

> The delay in starting or stopping your program depends on the quality of your network connection.
> Clicking the **Play** button converts your program to Python and sends it to the robot for execution.

To move back the robot to its default position. You can click on **Home Position** button.

Open an existing program

You can open an existing program by clicking the **Open** button on the toolbar. NiryoStudio will prompt you to select a program to open. Click on the entry to load it.



Import or Export a program

You can import a program by clicking the **Open** button on the toolbar, then selecting the **Import** button. Choose a Blockly program file, and NiryoStudio expects a *JSON* file.
You can export a program by clicking the **Export** button on the toolbar. Blockly programs are exported in *JSON* format.

Adding a position or a trajectory to your program

There are two ways to add a position or a trajectory: through the NiryoStudio interface or by using the buttons on the control panel.

Using NiryoStudio UI

You can add a position or trajectory block based on the robot's current state by clicking the **Add Position or Trajectory Block** button 1 .

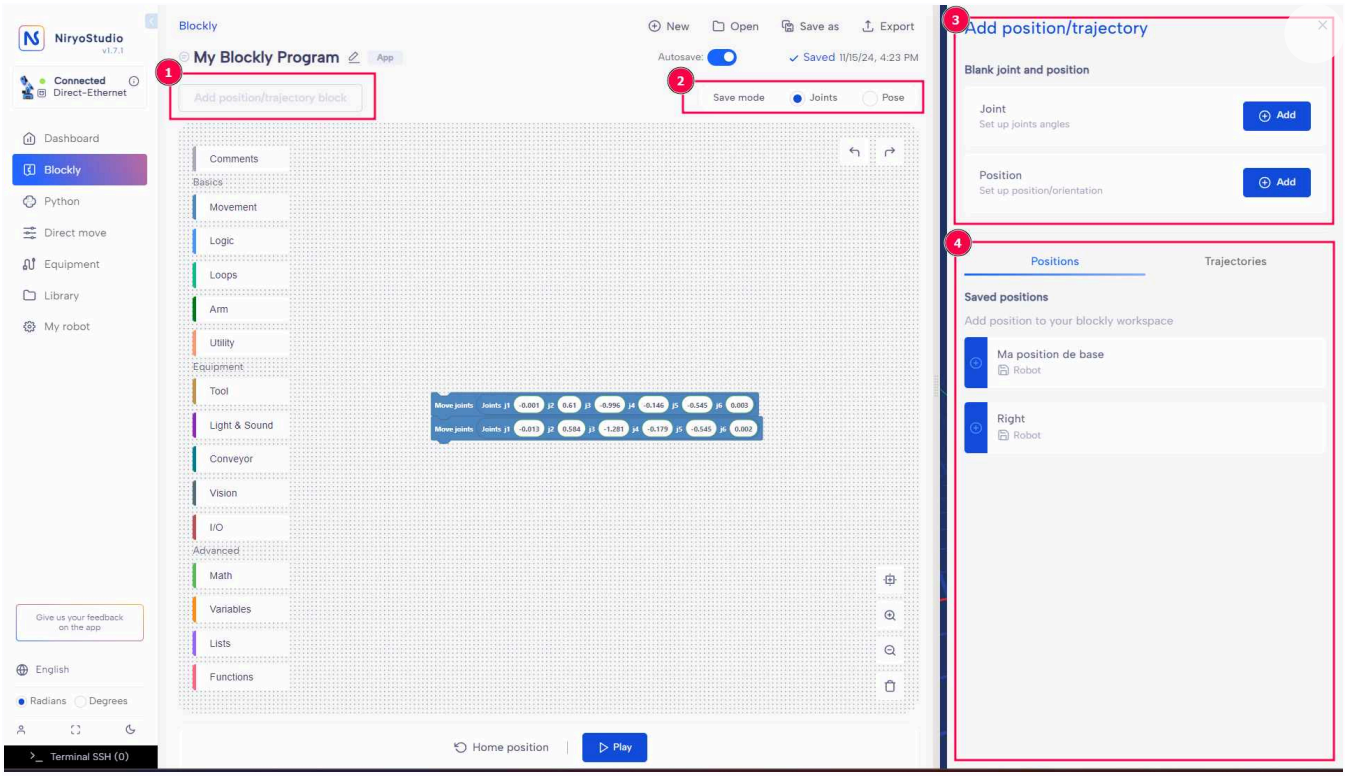
A slide panel will open, allowing you to choose between **Add Pose** or **Add Joint** 3 .

- **Add Pose**: Creates a new position block using position values (x, y, z, roll, pitch, yaw).
- **Add Joint**: Creates a new position block using joint values (J1, J2, J3, ...).

These values are taken from the robot's current position.

You can import a previously saved position or trajectory 4 . For more details on saved positions, refer to the

> Direct Move section
> https://docs.niryo.com/niryostudio/interface/direct-move/



> The save mode defines the type of block that will be created when you click the robot's **Save** button on End Effector. See next section.

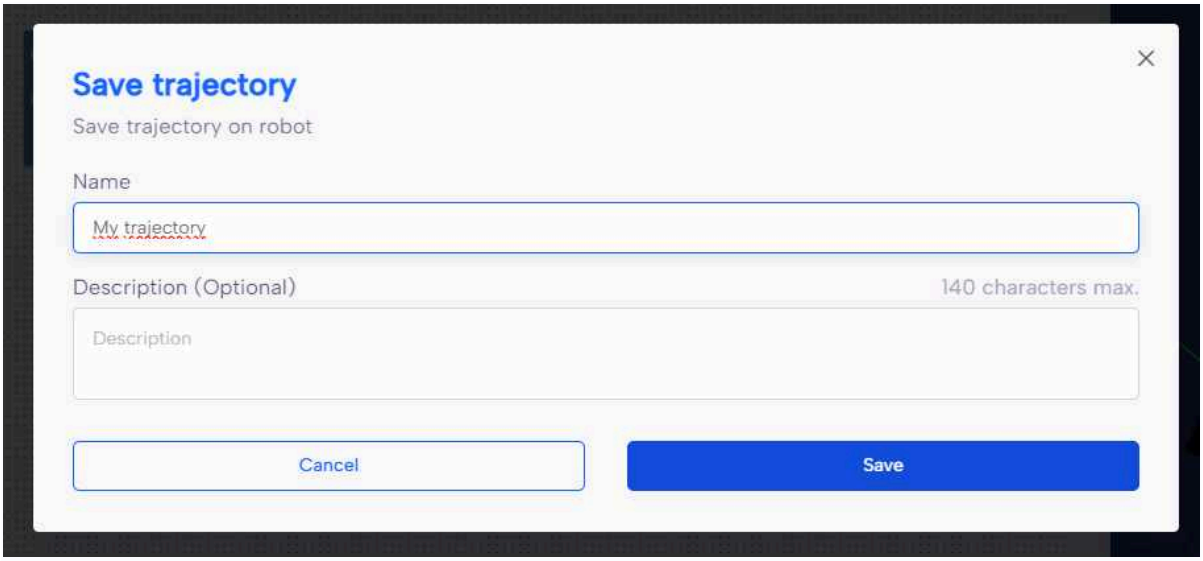
Using Robot End Effector

From the robot, you can save its position by pressing the **Save** button on the End Effector. A **Joint** or **Pose** block corresponding to the robot's position is then added to the program based

on the save configuration [2] .



To record a trajectory, you must hold down the **Save** button without releasing it. While holding the button, you can move the robot freely without resistance. Once the **Save** button is released, NiryoStudio displays a trajectory save popup and saves the trajectory to the robot.



> Note that trajectories are stored locally on the robot.
> You need to connect to SSH to the robot and go to
> *$HOME/niryo_robot_saved_files/niryo_robot_trajectories*

Blockly API

To see the documentation for all the block available and start coding with Blockly, please check the Blockly API

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